

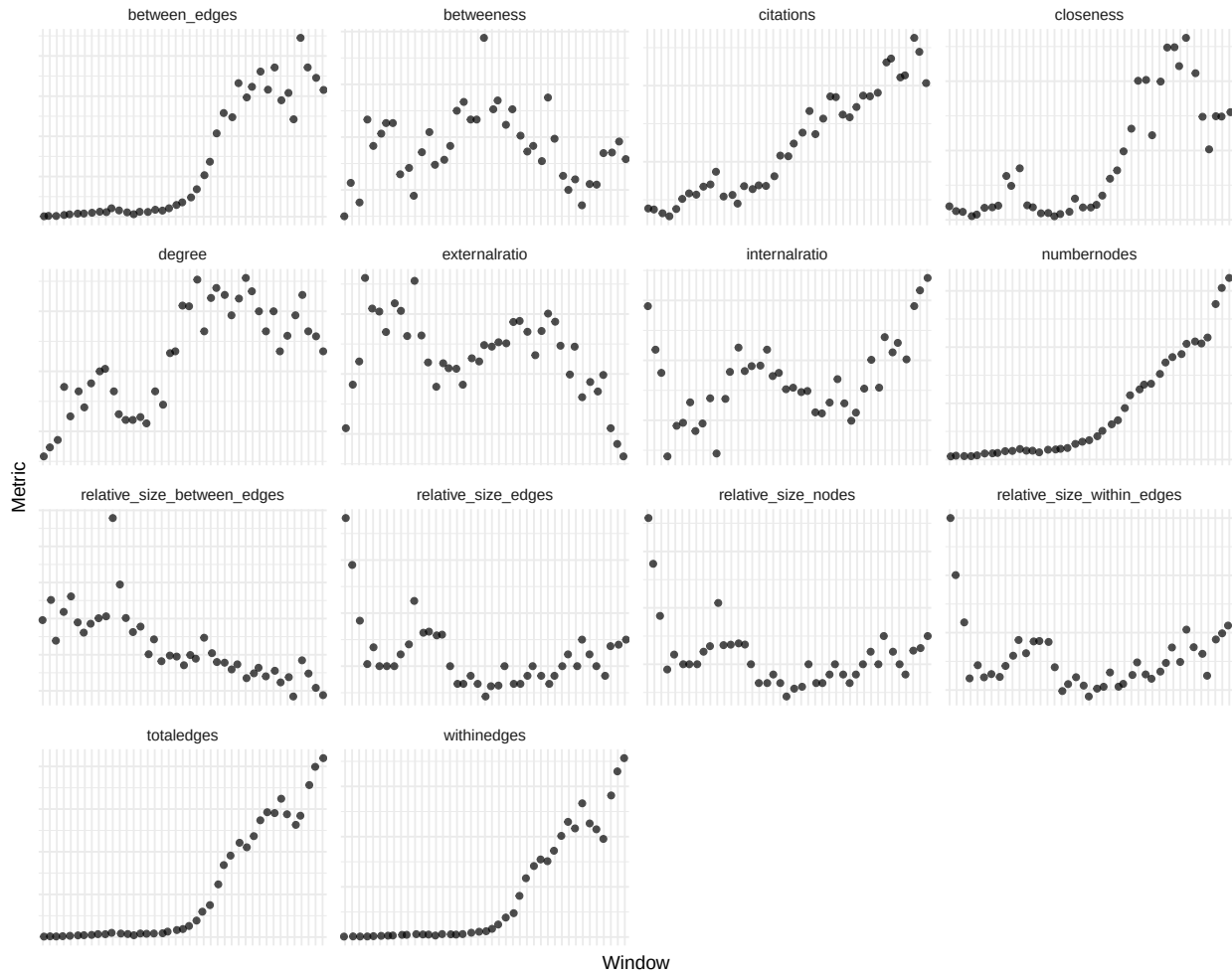
Cluster-Level Analysis

2025-11-27

1. Aggregate Metrics

The Metrics used here are formally defined on a per-field basis below. The aggregate metrics are computed similarly, but over all fields instead of per field.

1.1 Aggregate Evolution of Metrics



1.2 Number of Nodes and Edges over Time

Window	Number of Nodes	Number of Edges
1980-1984	49	120
1981-1985	73	217
1982-1986	86	253
1983-1987	127	499
1984-1988	137	623
1985-1989	216	986
1986-1990	221	1091
1987-1991	235	1270
1988-1992	269	1535
1989-1993	265	1400
1990-1994	243	1540
1991-1995	270	1572
1992-1996	269	1341
1993-1997	217	835
1994-1998	298	1607
1995-1999	359	1945
1996-2000	458	2516
1997-2001	494	2683
1998-2002	611	3513
1999-2003	761	4935
2000-2004	964	6577
2001-2005	1058	8007
2002-2006	1275	11822
2003-2007	1254	14875
2004-2008	1678	22471
2005-2009	2197	37060
2006-2010	2515	46415
2007-2011	2496	47739
2008-2012	2932	60787
2009-2013	3236	63140
2010-2014	3351	65068
2011-2015	3453	68445
2012-2016	3280	65846
2013-2017	3752	72667
2014-2018	3294	64860
2015-2019	3780	64776
2016-2020	4129	65765
2017-2021	4775	78229
2018-2022	4936	74938
2019-2023	5340	82963
2020-2024	5168	83887

1.3 Cluster Descriptives

Table 1: Cluster Descriptives

ID	Top Words	Window (Start)	Window (End)	Cluster Length	Max. Cluster Share
cl_1	voting schemes, voters, dynamic contract	1980-1984	1982-1986	3	22.54
cl_3	generic strategies, market share, attraction models	1980-1984	1984-1988	5	4.23
cl_4	evolutionary, prey, sperm	1980-1984	1996-2000	17	25.88
cl_7	ntu, axiomatization, n.e.p	1980-1984	1984-1988	5	16.47
cl_15	decision costs, sidepayment, effectiveness	1981-1985	1983-1987	3	14.12
cl_21	policy, expectations, rational expectations	1982-1986	1986-1990	5	10.08
cl_28	coalition, coalitions, blocs	1982-1986	1984-1988	3	5.77
cl_33	arms, star wars, deterrence	1983-1987	1988-1992	6	13.30
cl_36	repeated, discounting, oligopoly	1983-1987	1985-1989	3	13.33
cl_37	games, information dependent, common knowledge	1983-1987	1997-2001	15	23.70
cl_53	entry, entry preventing, queuing rule	1984-1988	1987-1991	4	6.01
cl_61	stochastic games, stochastic, contraction assumption	1985-1989	1989-1993	5	7.73
cl_64	games na, strategic threats, utility normal	1985-1989	1989-1993	5	9.01
cl_68	uncovered set, participation constraints, institutions	1985-1989	1987-1991	3	5.56
cl_72	contracts, potential surplus, arms rivalry	1985-1989	1990-1994	6	13.25
cl_83	bang, repeated games, repeated	1986-1990	1991-1995	6	14.90
cl_99	repeated games, repeated, brand	1987-1991	1989-1993	3	7.41
cl_107	nonobservable, nonobservable actions, repeated games	1988-1992	1992-1996	5	4.23
cl_138	ramsey, ramsey policy, repeated games	1990-1994	1992-1996	3	7.95
cl_157	stochastic games, stochastic, pi	1991-1995	2002-2006	12	5.31
cl_168	nucleolus, sscl, cost allocation	1991-1995	1997-2001	7	4.85
cl_173	cod, arcto, arcto norwegian	1992-1996	1994-1998	3	5.78
cl_202	outsourcing, product, firms	1993-1997	1995-1999	3	5.87

cl_212	international, chambers, risky states	1993-1997	1997-2001	5	9.43
cl_224	supply chain, supply, manufacturer	1994-1998	2020-2024	27	12.32
cl_246	trust, bargaining, social	1995-1999	2006-2010	12	16.46
cl_254	users, network, routing	1995-1999	2002-2006	8	7.69
cl_263	unplanned coordination, genetic, market selection	1995-1999	2002-2006	8	4.18
cl_278	evolutionary, cooperation, evolutionary game	1996-2000	2020-2024	25	22.24
cl_282	repeated, repeated games, monitoring	1996-2000	2008-2012	13	7.38
cl_285	shapley, coalition, cooperative	1996-2000	2020-2024	25	7.73
cl_300	stochastic games, crowding game, crowding	1997-2001	2000-2004	4	4.34
cl_318	fuzzy, linguistic fuzzy, logic	1998-2002	2006-2010	9	11.43
cl_323	power indices, rationalizability, de	1998-2002	2000-2004	3	5.85
cl_365	power control, power, control	1999-2003	2010-2014	12	5.63
cl_382	auctions, combinatorial auctions, combinatorial	2000-2004	2002-2006	3	4.22
cl_396	nodes, ad hoc, hoc	2000-2004	2009-2013	10	4.54
cl_496	network, games, congestion	2003-2007	2010-2014	8	9.59
cl_498	bargaining, negotiation, ghg	2003-2007	2005-2009	3	4.81
cl_535	spectrum, interference, users	2004-2008	2017-2021	14	12.17
cl_555	learning, electricity, reinforcement	2004-2008	2008-2012	5	4.64
cl_726	games, quantum, theory	2007-2011	2017-2021	11	11.88
cl_792	social, games, cooperation	2008-2012	2013-2017	6	8.47
cl_877	games, stochastic games, repeated	2009-2013	2013-2017	5	6.31
cl_891	learning, reward shaping, algorithm	2009-2013	2011-2015	3	4.57
cl_927	energy, grid, electricity	2009-2013	2020-2024	12	6.53
cl_1052	data, users, cloud	2011-2015	2017-2021	7	5.01
cl_1212	defense, attack, security	2013-2017	2020-2024	8	7.78
cl_1399	offloading, edge, edge computing	2015-2019	2020-2024	6	6.05
cl_1476	learning, reinforcement learning, reinforcement	2016-2020	2020-2024	5	6.50
cl_1545	evolutionary, government, enterprises	2017-2021	2020-2024	4	15.84

1.4 Top Articles per Cluster

Table 2: Top Articles per Cluster

ID	Top Words	Title	Authors	Year	Citations
cl_1	voting schemes, voters, dynamic contract	Reputation and imperfect information	David M. Kreps;Robert Wilson	1982	3101
		The Folk Theorem in Repeated Games with Discounting or with Incomplete Information	Drew Fudenberg;Eric Maskin	1986	2371
		Games of perfect information, predatory pricing and the chain-store paradox	Robert W. Rosenthal	1981	764
cl_1052	data, users, cloud	Distributed Energy Trading in Microgrids: A Game-Theoretic Model and Its Equilibrium Analysis	Joohyung Lee;Jun Guo;Jun Kyun Choi;Moshe Zukerman	2015	388
		Smart Cloud Storage Service Selection Based on Fuzzy Logic, Theory of Evidence and Game Theory	Christian Esposito;Massimo Ficco;Francesco Palmieri;Aniello Castiglione	2015	173
		Stochastic Games for Power Grid Protection Against Coordinated Cyber-Physical Attacks	Longfei Wei;Arif I. Sarwat;Walid Saad;Saroj Biswas	2016	161
cl_107	nonobservable, nonobservable actions, repeated games	Coalition-Proof Equilibrium	Diego Moreno;John Wooders	1996	124
		Existence of Stationary Correlated Equilibria with Symmetric Information for Discounted Stochastic Games	Andrzej S. Nowak;T. E. S. Raghavan	1992	109
		The Existence of Subgame-Perfect Equilibrium in Continuous Games with Almost Perfect Information: A Case for Public Randomization	Christopher Harris;Philip J. Reny;Arthur J. Robson	1995	89
cl_1212	defense, attack, security	Multi-Agent Reinforcement Learning-Based Resource Allocation for UAV Networks	Jingjing Cui;Yuanwei Liu;Arumugam Nallanathan	2019	498

		Deep Multi-User Reinforcement Learning for Distributed Dynamic Spectrum Access	Oshri Naparstek;Kobi Cohen	2018	443
		Resilient Control of Networked Control System Under DoS Attacks: A Unified Game Approach	Yuan Yuan;Huanhuan Yuan;Lei Guo;Hongjiu Yang;Shanlin Sun	2016	319
cl_138	ramsey, ramsey policy, repeated games	Non-computable strategies and discounted repeated games	John Nachbar;William R. Zame	1996	50
		Optimality and domination in repeated games with bounded players	Lance Fortnow;Duke Whang	1994	27
		Bounded rationality, neural network and folk theorem in repeated games with discounting	In-Koo Cho	1994	24
cl_1399	offloading, edge, edge computing	Mobile Edge Computing Enabled 5G Health Monitoring for Internet of Medical Things: A Decentralized Game Theoretic Approach	Zhaolong Ning;Peiran Dong;Xiaojie Wang;Xiping Hu;Lei Guo;Bin Hu;Yi Guo;Tie Qiu;Ricky Y. K. Kwok	2020	367
		A Game-Theoretical Approach for User Allocation in Edge Computing Environment	Qiang He;Guangming Cui;Xuyun Zhang;Feifei Chen;Shuiguang Deng;Hai Jin;Yanhui Li;Yun Yang	2019	354
		Online Distributed Offloading and Computing Resource Management With Energy Harvesting for Heterogeneous MEC-Enabled IoT	Shichao Xia;Zhixiu Yao;Yun Li;Shiwen Mao	2021	255
cl_1476	learning, reinforcement learning, reinforcement	Deep Reinforcement Learning for Cyber Security	Thanh Thi Nguyen;Vijay Janapa Reddi	2021	424
		Generative Adversarial Networks	Zhipeng Cai;Zuobin Xiong;Honghui Xu;Peng Wang;Wei Li;Yi Pan	2021	325
		Multi-Objective Workflow Scheduling With Deep-Q-Network-Based Multi-Agent Reinforcement Learning	Yuandou Wang;Hang Liu;Wanbo Zheng;Yunni Xia;Yawen Li;Peng Chen;Kunyin Guo;Hong Xie	2019	271

cl_15	decision costs, sidepayment, effectiveness	Ambiguity and accounting: The elusive link between information and decision making	James G. March	1987	343
		What can managers do about unethical management?	Richard P. Nielsen	1987	37
		Playing doctor: Application of game theory to medical decision-making	George Diamond;Alan Rozanski;Michael Steuer	1986	32
cl_1545	evolutionary, government, enterprises	Government regulation to promote coordinated emission reduction among enterprises in the green supply chain based on evolutionary game analysis	Zheng Liu;Qingshan Qian;Bin Hu;Wen-Long Shang;Lingling Li;Yuanjun Zhao;Zhao Zhao;Chunjia Han	2022	197
		Implementation of enterprises' green technology innovation under market-based environmental regulation: An evolutionary game approach	Mengyuan Li;Xin Gao	2022	186
		Evolutionary game analysis on behavior strategies of multiple stakeholders in maritime shore power system	Lang Xu;Zhongjie Di;Jihong Chen;Jia Shi;Yang Chen	2021	174
cl_157	stochastic games, stochastic, pi	Decentralized learning of Nash equilibria in multi-person stochastic games with incomplete information	P. S. Sastry;V.V. Phansalkar;M.A.L. Thathachar	1994	318
		Regret Minimization Under Partial Monitoring	Nicolò Cesa-Bianchi;Gábor Lugosi;Gilles Stoltz	2006	145
		Oligopoly games with nonlinear demand and cost functions: Two boundedly rational adjustment processes	Ahmad Naimzada;Lucia Sbragia	2005	120
cl_168	nucleolus, sscl, cost allocation	The Core of an Economy with Multilateral Environmental Externalities	Parkash Chander;Henry Tulkens	1997	421
		The core of an economy with multilateral environmental externalities	Parkash Chander;Henry Tulkens	1997	311

		Applications of approximation algorithms to cooperative games	Kamal Jain;Vijay V. Vazirani	2001	210
cl_173	cod, arcto, arcto norwegian	Social conflict and growth	Jess Benhabib;Aldo Rustichini	1996	423
		Where are we in the political economy of reform?	Mariano Tommasi;Andrés Velasco	1996	204
		Existence and Properties of Equilibria for a Dynamic Game on Productive Assets	Engelbert J. Dockner;Gerhard Sorger	1996	103
cl_202	outsourcing, product, firms	The Interorganizational Learning Dilemma: Collective Knowledge Development in Strategic Alliances	Rikard Larsson;Lars Bengtsson;Kristina Henriksson;Judith Sparks	1998	1077
		Competing in Product and Service: A Product Life-Cycle Model	Morris A. Cohen;Seungjin Whang	1997	218
		Marketing Information: A Competitive Analysis	Miklós Sárváry;Philip M. Parker	1997	106
cl_21	policy, expectations, rational expectations	Credibility and Policy Convergence in a Two-party System with Rational Voters	Alberto Alesina	1988	823
		The Effects of Brand Loyalty on Competitive Price Promotional Strategies	Jagmohan S. Raju;V. Srinivasan;Rajiv Lal	1990	368
		Price Promotions: Limiting Competitive Encroachment	Rajiv Lal	1990	170
cl_212	international, chambers, risky states	Economic Analysis of Social Interactions	Charles F. Manski	2000	2052
		Bargaining and Distribution in Marriage	Shelly Lundberg;Robert A. Pollak	1996	1143
		What's in a Name? Reputation as a Tradeable Asset	Steven Tadelis	1999	451
cl_224	supply chain, supply, manufacturer	Imitation of Complex Strategies	Jan W. Rivkin	2000	1292
		Supply chain analysis under green sensitive consumer demand and cost sharing contract	Debabrata Ghosh;Janat Shah	2014	773
		Building Store Loyalty through Store Brands	Marcel Corstjens;Rajiv Lal	2000	667
cl_246	trust, bargaining, social	Maximizing the spread of influence through a social network	David Kempe;Jon Kleinberg;Éva Tardos	2003	7088

		A theory of sequential reciprocity	Martin Dufwenberg;Georg Kirchsteiger	2003	1902
		Unraveling in Guessing Games: An Experimental Study	Rosemarie Nagel	2007	1483
cl_254	users, network, routing	Truth revelation in approximately efficient combinatorial auctions	Daniel Lehmann;Liadan O'Callaghan;Yoav Shoham	2002	653
		Selfish Routing in Capacitated Networks	José Correa;Andreas S. Schulz;Nicolás E. Stier-Moses	2004	386
		Bounding the inefficiency of equilibria in nonatomic congestion games	Tim Roughgarden;Éva Tardos	2003	298
cl_263	unplanned coordination, genetic, market selection	Job Matching and the Wage Distribution	Giuseppe Moscarini	2005	215
		Does chaos work better than noise?	Maide Bucolo;Riccardo Caponetto;Luigi Fortuna;Mattia Frasca;Alessandro Rizzo	2002	168
		Evolution of Reinforcement Learning in Uncertain Environments: A Simple Explanation for Complex Foraging Behaviors	Y. Niv;Daphna Joel;Isaac Meilijson;Eytan Ruppin	2002	158
cl_278	evolutionary, cooperation, evolutionary game	Evolutionary games on graphs	György Szabó;Gábor Fáth	2007	2823
		A simple rule for the evolution of cooperation on graphs and social networks	Hisashi Ohtsuki;Christoph Hauert;E. James Lieberman;Martin A. Nowak	2006	2155
		Emergence of cooperation and evolutionary stability in finite populations	Martin A. Nowak;Akira Sasaki;Christine Taylor;Drew Fudenberg	2004	1378
cl_28	coalition, coalitions, blocs	Size of smallest coalition as a source of power in coalition bargaining	David A. Kravitz	1987	18
		Coalition behavior: Effects of earned versus unearned resources	Charles E. Miller;Jane Yeang Chui Wong	1986	9
cl_282	repeated, repeated games, monitoring	Which Side Are You On? Bias, Credibility, and Mediation	Andrew Kydd	2003	385
		Violence Against Civilians in Civil Wars: Looting or Terror?	Jean-Paul Azam;Anke Hoeffler	2002	309

		A game-theoretic model for generation expansion planning: problem formulation and numerical comparisons	A.S. Chuang;FF Wu;Pravin Varaiya	2001	229
cl_285	shapley, coalition, cooperative	A bargaining perspective on resource advantage	Steven A. Lippman;Richard P. Rumelt	2003	600
		Axiomatic and game-theoretic analysis of bankruptcy and taxation problems: a survey	William Thomson	2003	598
		Biform Games	Adam Brandenburger;Harborne W. Stuart	2007	324
cl_3	generic strategies, market share, attraction models	Generic competitive strategies—An analytical approach	Aneel Karnani	1984	184
		Strategic Implications of Market Share Attraction Models	Aneel Karnani	1985	64
		The Value of Market Share and the Product Life Cycle—A Game-Theoretic Model	Aneel Karnani	1984	28
cl_300	stochastic games, crowding game, crowding	Efficient power control via pricing in wireless data networks	Cem Saraydar;Narayan B. Mandayam;D.J. Goodman	2002	1332
		Distributed multiuser power control for digital subscriber lines	Wei Yu;G. Ginis;J.M. Cioffi	2002	866
		Power control for wireless data	D.J. Goodman;Narayan B. Mandayam	2000	652
cl_318	fuzzy, linguistic fuzzy, logic	Game-theoretic analysis of cooperation among supply chain agents: Review and extensions	Mahesh Nagarajan;Greys Sošić	2006	607
		Introduction to game theory	Shengkai Zhang;Yadong Zhang	2003	512
		A Modal Logic for Coalitional Power in Games	Marc Pauly	2002	466
cl_323	power indices, rationalizability, de	Multi-agent influence diagrams for representing and solving games	Daphne Koller;Brian Milch	2003	399
		Rationalization and Incomplete Information	Pierpaolo Battigalli;Marciano Siniscalchi	2003	177
		Dynamic interactive epistemology	Oliver Board	2004	152

cl_33	arms, star wars, deterrence	Nash equilibrium with strategic complementarities	Xavier Vives	1990	921
		An Outside Option Experiment	Ken Binmore;Avner Shaked;John Sutton	1989	369
		Game theoretic analysis of the substitutable product inventory problem with random demands	Mahmut Parlar	1988	352
cl_36	repeated, discounting, oligopoly	A Theory of Dynamic Oligopoly, II: Price Competition, Kinked Demand Curves, and Edgeworth Cycles	Eric Maskin;Jean Tirole	1988	720
		The Structure of Nash Equilibrium in Repeated Games with Finite Automata	Dilip Abreu;Ariel Rubinstein	1988	391
		Extensive Form Games in Continuous Time: Pure Strategies	Leo K. Simon;Maxwell B. Stinchcombe	1989	252
cl_365	power control, power, control	Spectrum sharing for unlicensed bands	Raúl Etkin;Abhay Parekh;David Tse	2007	879
		Distributed Relay Selection and Power Control for Multiuser Cooperative Communication Networks Using Stackelberg Game	Beibei Wang;Zhu Han;K. J. Ray Liu	2008	414
		Individual and mass behaviour in large population stochastic wireless power control problems: centralized and nash equilibrium solutions	Minyi Huang;P.E. Caines;Roland P. Malhamé	2004	364
cl_37	games, information dependent, common knowledge	Rationalizability, Learning, and Equilibrium in Games with Strategic Complementarities	Paul Milgrom;John Roberts	1990	1816
		Correlated Equilibrium as an Expression of Bayesian Rationality	Robert J. Aumann	1987	1603
		Psychological games and sequential rationality	John Geanakoplos;David Pearce;Ennio Stacchetti	1989	1034
cl_382	auctions, combinatorial auctions, combinatorial	Computing the optimal strategy to commit to	Vincent Conitzer;Tüomas Sandholm	2006	475
		Game strategies in network security	Kong-wei Lye;Jeannette M. Wing	2005	433

		Incentive-based modeling and inference of attacker intent, objectives, and strategies	Peng Liu;Wanyu Zang;Meng Yu	2005	200
cl_396	nodes, ad hoc, hoc	Using game theory to analyze wireless ad hoc networks	Vivek Srivastava;James Neel;Allen B. MacKenzie;R. Menon;Luiz A. DaSilva;J. Hicks;Jeffrey H. Reed;Robert P. Gilles	2005	477
		Pseudonym Changing at Social Spots: An Effective Strategy for Location Privacy in VANETs	Rongxing Lu;Xiaodong Lin;Tom H. Luan;Xiaohui Liang;Xuemin Shen	2011	454
		Adaptive channel allocation spectrum etiquette for cognitive radio networks	Nie Nie;Cristina Comaniciu	2005	426
cl_4	evolutionary, prey, sperm	Evolutionary Games in Economics	Daniel Friedman	1991	1797
		Alternative reproductive strategies and tactics: diversity within sexes	Mart R. Gross	1996	1681
		Rational Choice and Turnout	John H. Aldrich	1993	1210
cl_496	network, games, congestion	Decentralized Charging Control of Large Populations of Plug-in Electric Vehicles	Zhongjing Ma;Duncan S. Callaway;Ian A. Hiskens	2011	963
		The Complexity of Computing a Nash Equilibrium	Constantinos Daskalakis;Paul W. Goldberg;Christos H. Papadimitriou	2009	921
		Decentralized Computation Offloading Game for Mobile Cloud Computing	Xu Chen;Xu Chen	2014	843
cl_498	bargaining, negotiation, ghg	Quality Improvement Incentives and Product Recall Cost Sharing Contracts	Gary H. Chao;Seyed M. R. Iravani;R. Canan Savaskan	2009	358
		Co-opetition and Investment for Supply-Chain Resilience	Nitin Bakshi;Paul R. Kleindorfer	2009	232
		Quality investment and inspection policy in a supplier-manufacturer supply chain	Chung-Chi Hsieh;Yu-Te Liu	2009	124
cl_53	entry, entry preventing, queuing rule	Price Competition in a Channel Structure with a Common Retailer	Sungchul Choi	1991	1034

		Modeling, game theory, and strategic management	Garth Saloner	1991	124
		A Theory of Negotiated Equity Financing	Ronald M. Giammarino; Tracy R. Lewis	1988	66
cl_535	spectrum, interference, users	Autonomous Demand-Side Management Based on Game-Theoretic Energy Consumption Scheduling for the Future Smart Grid	Amir-Hamed Mohsenian-Rad; Vincent W. S. Wong; Juri Jatskevich; Robert Schober; Alberto Leon-Garcia	2010	2711
		Competitive Pricing for Spectrum Sharing in Cognitive Radio Networks: Dynamic Game, Inefficiency of Nash Equilibrium, and Collusion	Dusit Niyato; Ekram Hossain	2008	521
		Dynamics of Network Selection in Heterogeneous Wireless Networks: An Evolutionary Game Approach	Dusit Niyato; Ekram Hossain	2008	470
cl_555	learning, electricity, reinforcement	An anti-jamming stochastic game for cognitive radio networks	B Wang; Yongle Wu; K J R Liu; T. Charles Clancy	2011	315
		Directional routing and scheduling for green vehicular delay tolerant networks	Yuanyuan Zeng; Kai Xiang; Deshi Li; Athanasios V. Vasilakos	2012	300
		FlipIt: The Game of “Stealthy Takeover”	Marten van Dijk; Ari Juels; Alina Oprea; Ronald L. Rivest	2012	271
cl_61	stochastic games, stochastic, contraction assumption	Algorithms for stochastic games ? A survey	T. E. S. Raghavan; Jerzy A. Filar	1991	165
		On Repeated Games with Complete Information	Sylvain Sorin	1986	129
		Nonlinear programming and stationary equilibria in stochastic games	Jerzy A. Filar; Todd A. Schultz; Frank Thuijsman; O. J. Vrieze	1991	77
cl_64	games na, strategic threats, utility normal	The Shapley value for cooperative games under precedence constraints	Ulrich Faigle; Walter Kern	1992	252
		Games with permission structures: The conjunctive approach	Robert P. Gilles; Guillermo Owen; René van den Brink	1992	234
		Bargaining in the transfer market: theory and evidence	Fiona Carmichael; D Thomas	1993	93

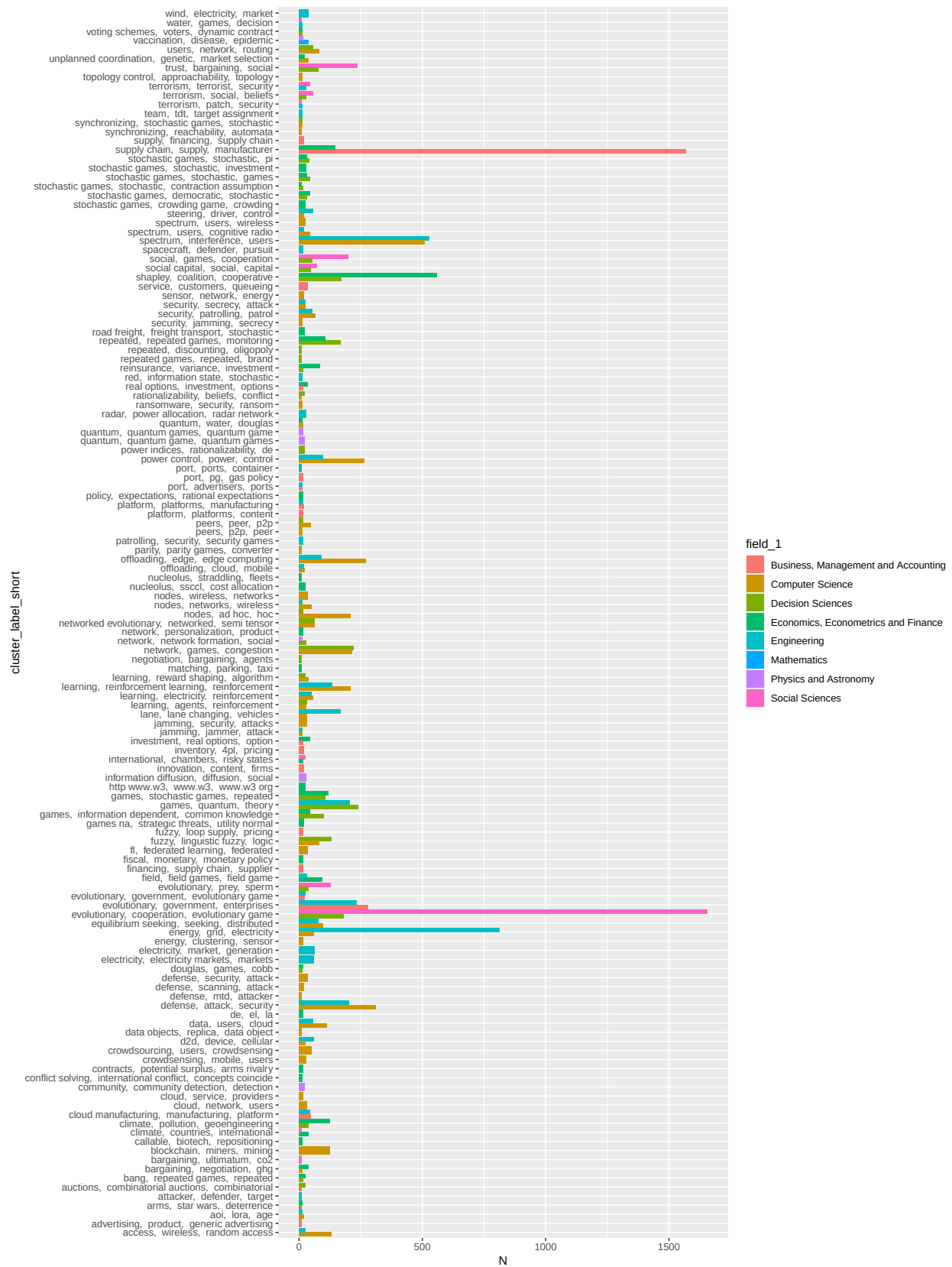
cl_68	uncovered set, participation constraints, institutions	A Concept of Egalitarianism Under Participation Constraints	Bhaskar Dutta;Debraj Ray	1989	357
		Nontransitive preferences in decision theory	Peter C. Fishburn	1991	249
		Elite Interaction and Constitution Building in Consociational Democracies	George Tsebelis	1990	128
cl_7	ntu, axiomatization, n.e.p	An Axiomatization of the Non-Transferable Utility Value	Robert J. Aumann	1985	164
		Game theory and the tennessee valley authority	Philip D. Straffin;James P. Heaney	1981	125
		Distortion of Utilities and the Bargaining Problem	Joel Sobel	1981	108
cl_72	contracts, potential surplus, arms rivalry	STRATEGIC ALLIANCE STRUCTURING: A GAME THEORETIC AND TRANSACTION COST EXAMINATION OF INTERFIRM COOPERATION.	Arvind Parkhe	1993	2581
		Implicit Contracts, Incentive Compatibility, and Involuntary Unemployment	W. Bentley MacLeod;James M. Malcomson	1989	782
		Reciprocity without commitment	Stephen Coate;Martin Ravallion	1993	651
cl_726	games, quantum, theory	A Monte-Carlo game theoretic approach for Multi-Criteria Decision Making under uncertainty	Kaveh Madani;Jay R. Lund	2011	188
		Sustainable energy, water and food nexus systems: A focused review of decision-making tools for efficient resource management and governance	Sarah Namany;Tareq Al-Ansari;Rajesh Govindan	2019	187
		Formulating an n-person noncooperative game as a tensor complementarity problem	Zheng-Hai Huang;Liqun Qi	2016	174
cl_792	social, games, cooperation	Structural Models of Nonequilibrium Strategic Thinking: Theory, Evidence, and Applications	Vincent P. Crawford;Miguel A. Costa-Gomes;Nagore Iriberri	2013	499

		Are groups more rational than individuals? A review of interactive decision making in groups	Tamar Kugler;Edgar E. Kausel;Martin G. Kocher	2012	308
		Industry or Academia, Basic or Applied? Career Choices and Earnings Trajectories of Scientists	Rajshree Agarwal;Atsushi Ohyama	2012	185
cl_83	bang, repeated games, repeated	Toward a Theory of Discounted Repeated Games with Imperfect Monitoring	Dilip Abreu;David Pearce;Ennio Stacchetti	1990	1245
		Dynamic voluntary provision of public goods	Chaim Fershtman;Shmuel Nitzan	1991	205
		Vancouver's Gasoline-Price Wars: An Empirical Exercise in Uncovering Supergame Strategies	Margaret E. Slade	1992	194
cl_877	games, stochastic games, repeated	Bounded Memory and Biases in Information Processing	Andrea C. Wilson	2014	160
		Strategy Iteration Is Strongly Polynomial for 2-Player Turn-Based Stochastic Games with a Constant Discount Factor	Thomas Dueholm Hansen;Peter Bro Miltersen;Uri Zwick	2013	128
		The Dynamic Efficiency Costs of Common-Pool Resource Exploitation	Ling Huang;Martin D. Smith	2014	119
cl_891	learning, reward shaping, algorithm	Jamming Attacks on Remote State Estimation in Cyber-Physical Systems: A Game-Theoretic Approach	Yuzhe Li;Ling Shi;Peng Cheng;Jiming Chen;Daniel E. Quevedo	2015	444
		Evolutionary Dynamics of Multi-Agent Learning: A Survey	Daan Bloembergen;Karl Tuyls;Daniel Hennes;Michael Kaisers	2015	273
		Game-Theoretic Analysis of the Hegselmann-Krause Model for Opinion Dynamics in Finite Dimensions	S. Rasoul Etesami;Tamer Başar	2015	176
cl_927	energy, grid, electricity	Peer-to-Peer Energy Trading in a Prosumer-Based Community Microgrid: A Game-Theoretic Model	Amrit Paudel;Kalpesh Chaudhari;Chao Long;Hoay Beng Gooi	2018	699

		Distributed Nash Equilibrium Seeking by a Consensus Based Approach	Maojiao Ye;Guoqiang Hu	2017	444
		A Game-Theoretic Approach to Energy Trading in the Smart Grid	Yunpeng Wang;Walid Saad;Zhu Han;H. Vincent Poor;Tamer Başar	2014	410
cl_99	repeated games, repeated, brand	Evolutionary stability in repeated games played by finite automata	Kenneth Binmore;Larry Samuelson	1992	369
		On the Value of Game Theory in Social Science	Roger B. Myerson	1992	59
		Finite automata equilibria with discounting	Michele Piccione	1992	36

1.5 Distribution of Clusters per Field (and vice versa)





2. Evolution of Clusters

Cluster-Level Metrics are derived from networks where articles are nodes and edges are based on the strength of bibliographic coupling between articles. The nodes, edges and their information is then aggregated per discipline, here called cluster.

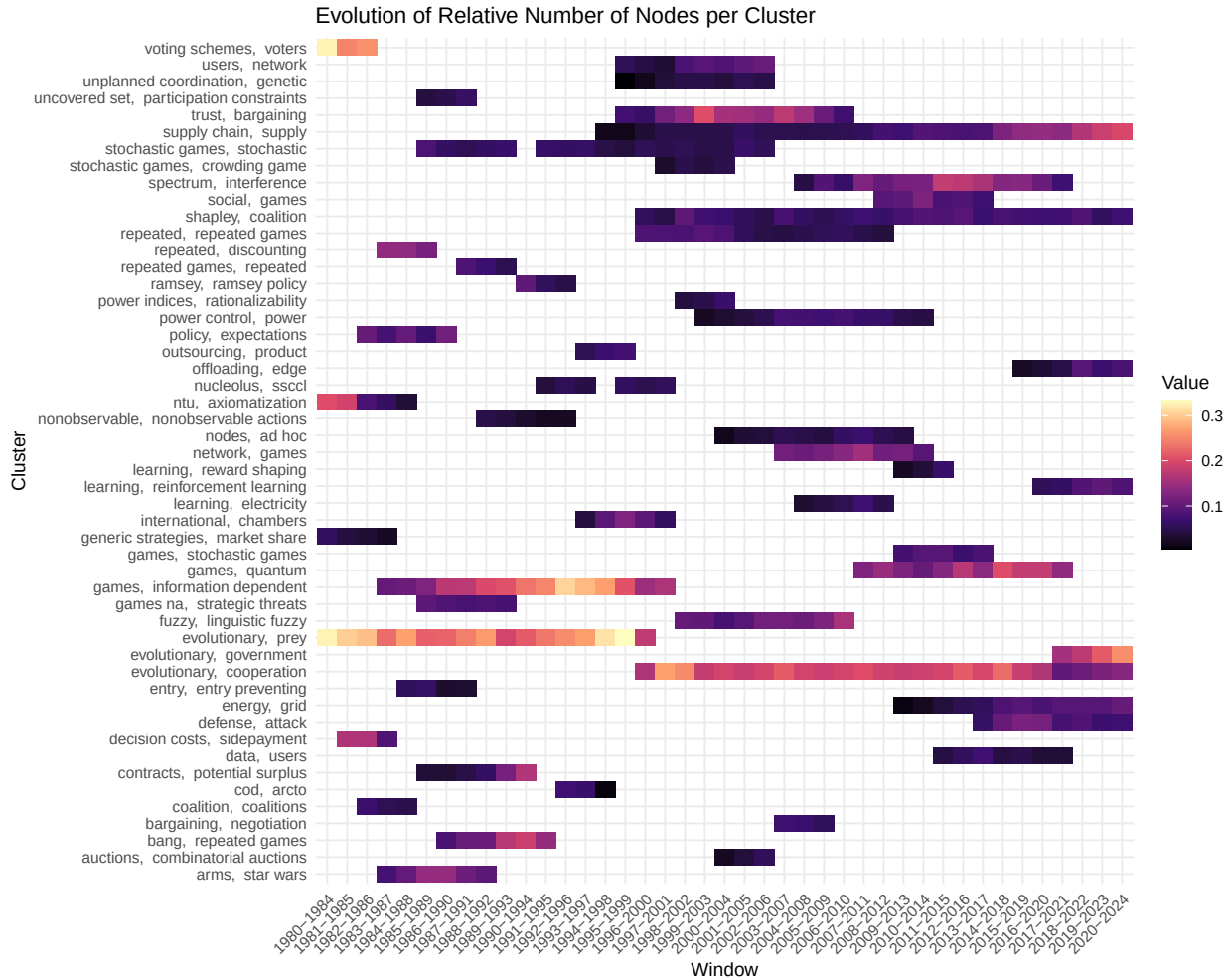
2. Relative Number of Nodes and Number of Citations

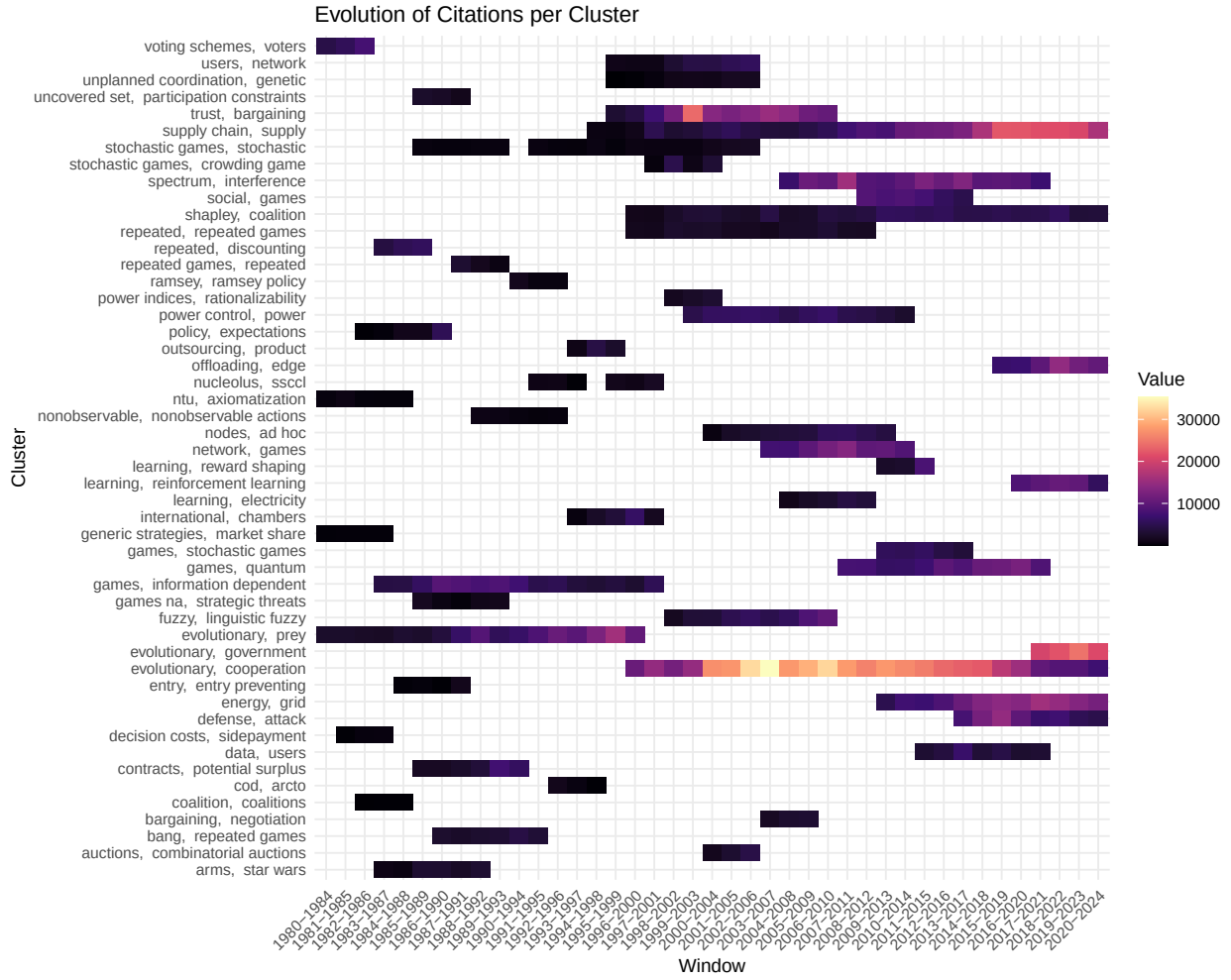
- Relative number of Nodes:

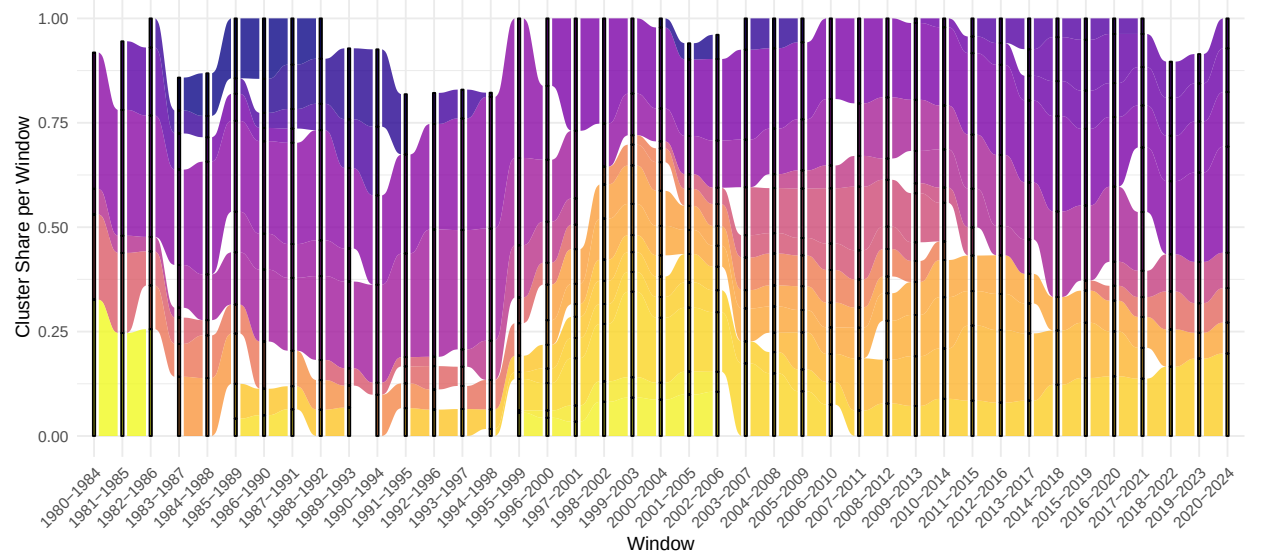
$$RelNodes_{c,f} = \frac{N(V)_{c,f}}{\sum_c N(V)_{c,f}} \quad (1)$$

where:

- $RelNodes_{t,c}$ measures the relative number of nodes for cluster c in time window t across all article nodes
- $N(V)_{t,c}$ measures the total number of nodes for cluster c in time window t across all article nodes. It is therefore defined as $N(V)_{c,f} = \sum_{v_i} 1_{cluster(v_i)=c}$, where $1_{cluster(v_i)=c}$ is an indicator function, taking the value one if article node v_i belongs to cluster c .
- The number of citations per cluster and time window is measured as the sum of citations across all articles of a cluster in a give time window, respectively.







ns, star wars	energy, grid	generic strategies, market share	offloading, edge	social, games
ctions, combinatorial auctions	entry, entry preventing	international, chambers	outsourcing, product	spectrum, interference
ng, repeated games	evolutionary, cooperation	learning, electricity	policy, expectations	stochastic games, cro
gaining, negotiation	evolutionary, government	learning, reinforcement learning	power control, power	stochastic games, sto
lition, coalitions	evolutionary, prey	learning, reward shaping	power indices, rationalizability	supply chain, supply
l, arcto	fuzzy, linguistic fuzzy	network, games	ramsey, ramsey policy	trust, bargaining
tracts, potential surplus	games na, strategic threats	nodes, ad hoc	repeated games, repeated	uncovered set, partici
a, users	games, information dependent	nonobservable, nonobservable actions	repeated, discounting	unplanned coordinatio
ision costs, sidepayment	games, quantum	ntu, axiomatization	repeated, repeated games	users, network
ense, attack	games, stochastic games	nucleolus, sscl	shapley, coalition	voting schemes, voter

2.2 Relative Number of Edges

- Relative Number of Total Edges:

$$RelEdges_{t,c} = \frac{N(E)_{t,c}}{\sum_f^F N(E)_{t,c}} \quad (2)$$

where:

- $RelEdges_{t,c}$ measures the relative number of edges incident on nodes within cluster c in time window t
- $N(E)_{t,c}$ measures the total number of edges incident on nodes within cluster c in time window t . It is therefore defined as $N(E)_{t,c} = \sum_{e=(v_i,v_j)} 1_{cluster(v_i)=c \vee cluster(v_j)=c}$, where $1_{cluster(v_k)=c}$ is an indicator function, taking the value one if article node k belongs to cluster c
- Relative Number of Within-cluster Edges:

$$RelEdgesWithin_{t,c} = \frac{N(E_{within})_{t,c}}{\sum_f^F N(E)_{t,c}} \quad (3)$$

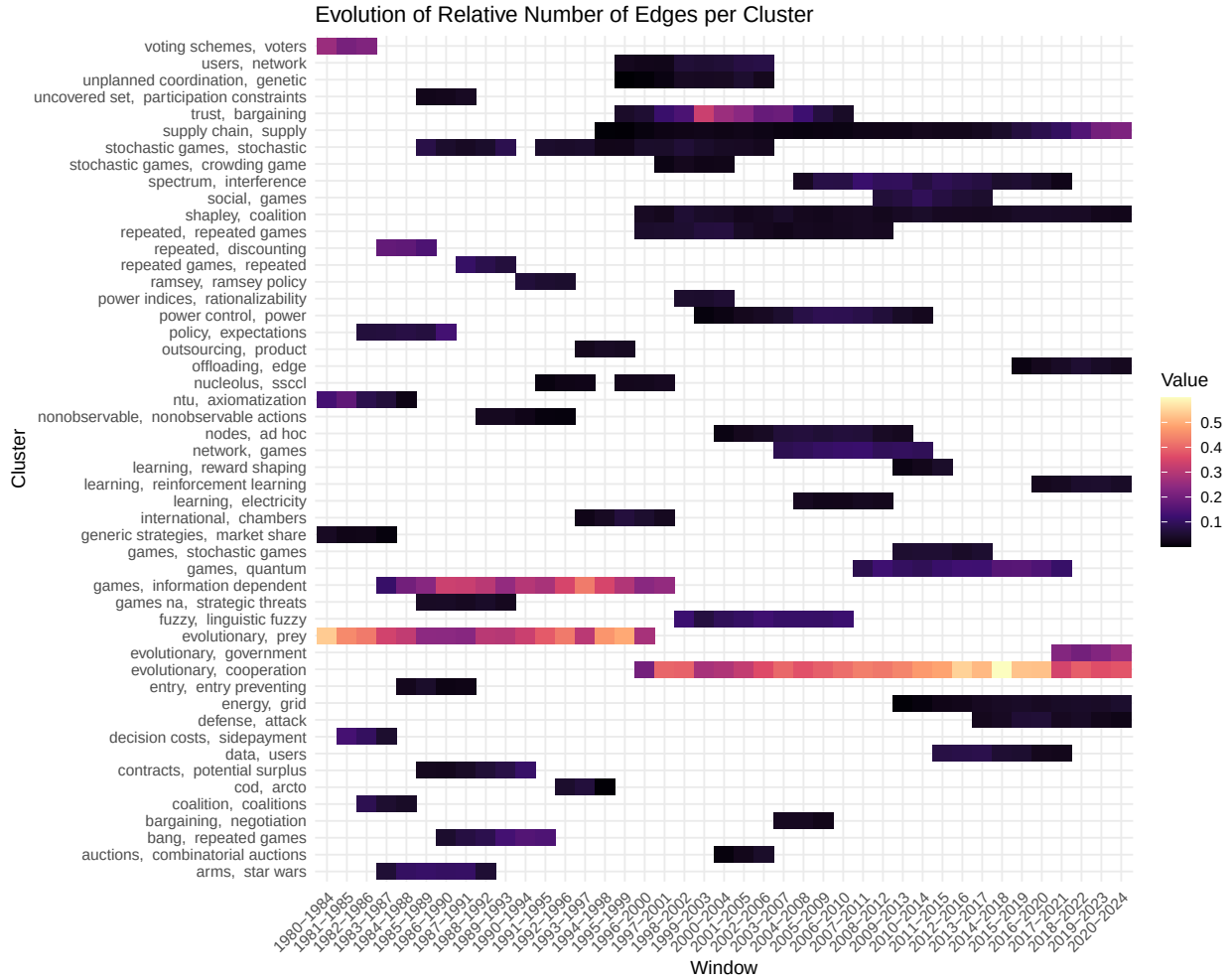
where:

- $RelEdgesWithin_{t,c}$ measures the relative number of edges between article nodes of the same cluster for cluster c in time window t
- $N(E_{within})_{t,c}$ measures the total number of edges between article nodes of the same cluster for cluster c in time window t . It is therefore defined as $N(E_{within})_{t,c} = \sum_{e=(v_i,v_j)} 1_{cluster\{(v_i)=c\}} 1_{\{cluster(v_j)=c\}}$, where $1_{cluster(v_k)=c}$ is an indicator function, taking the value one if article node k belongs to cluster c .
- Relative Number of Between-cluster Edges:

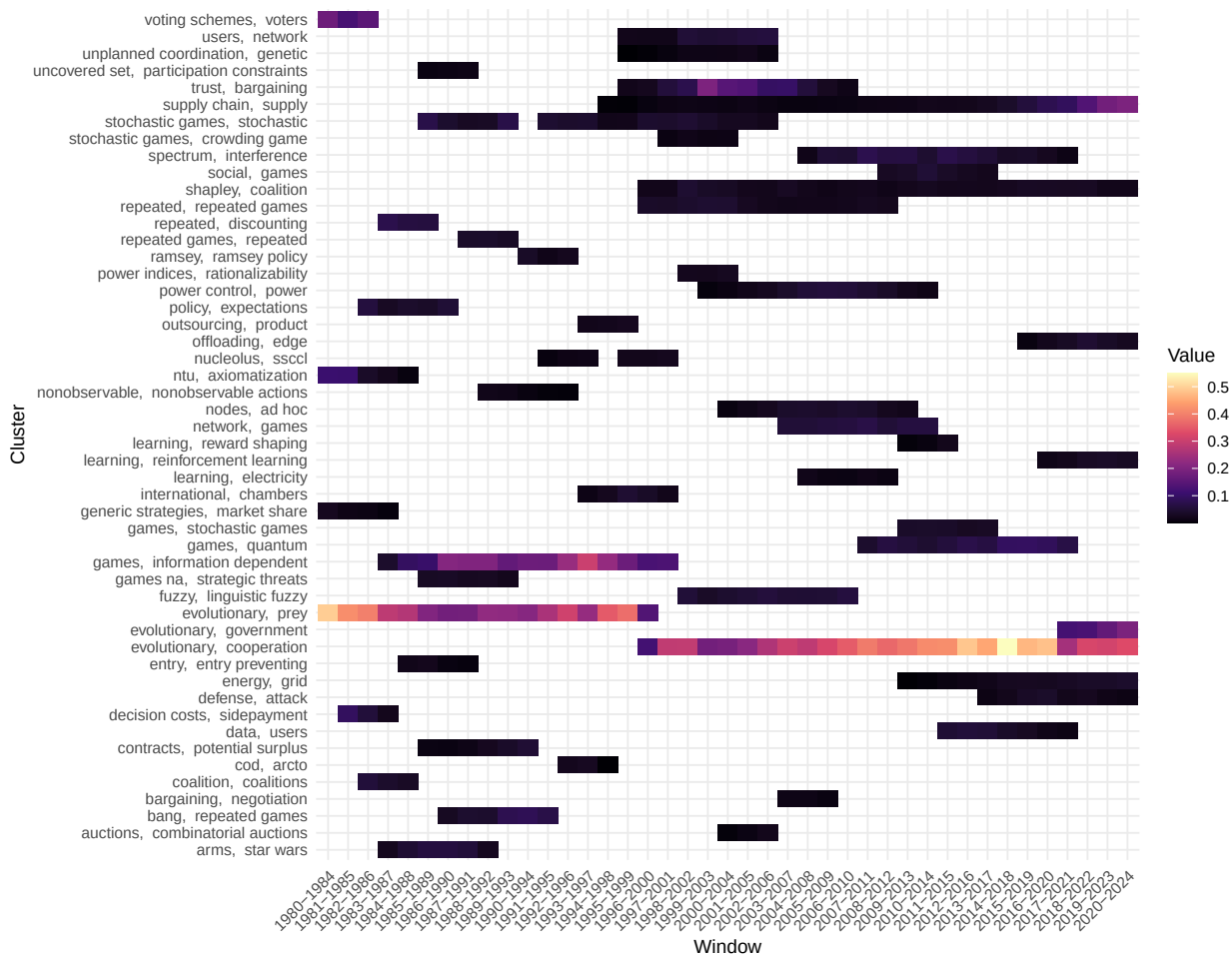
$$RelEdgesBetween_{t,c} = \frac{N(E_{Between})_{t,c}}{\sum_f^F N(E)_{t,c}} \quad (4)$$

where:

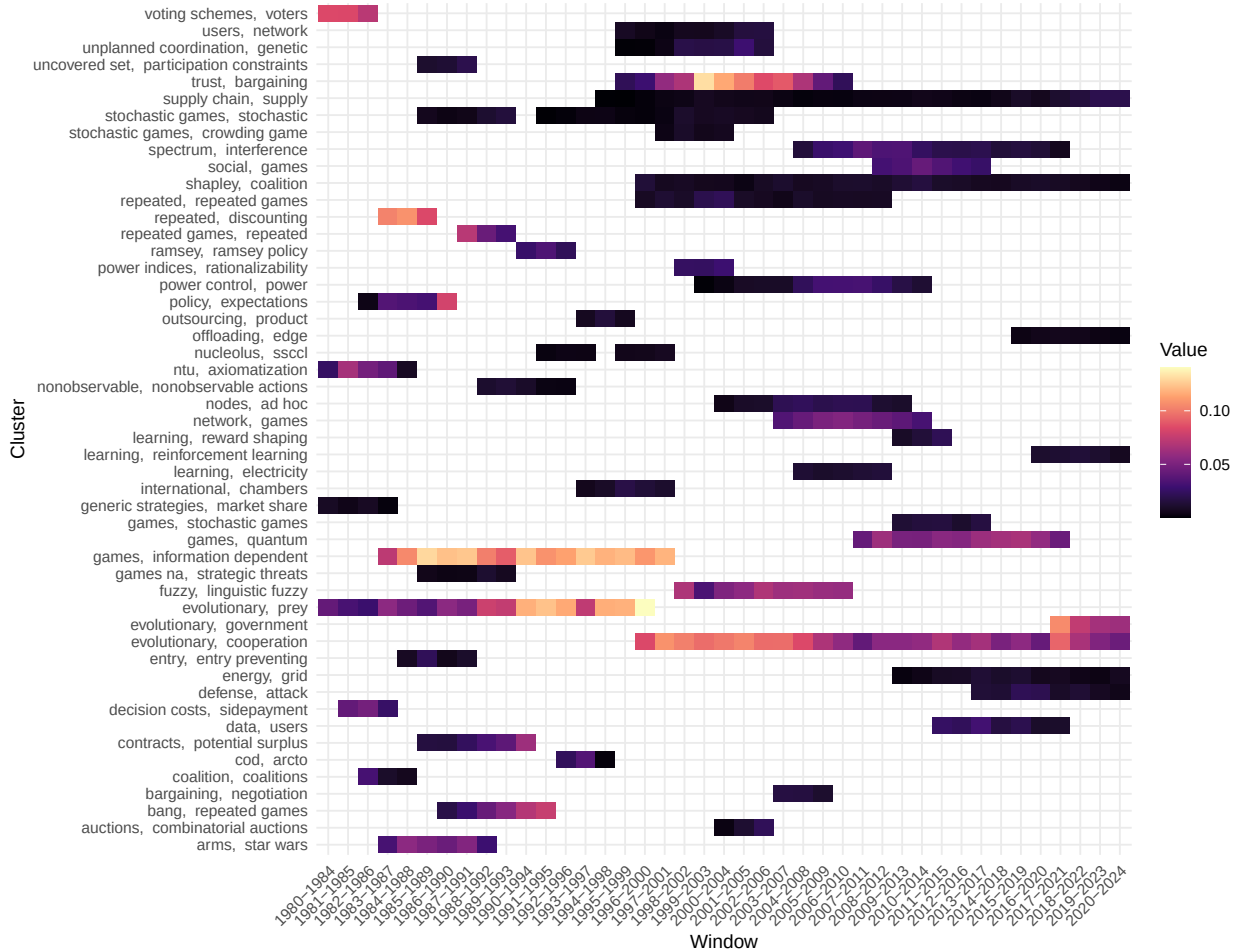
- $RelEdgesBetween_{t,c}$ measures the relative number of edges between article nodes of different clusters for cluster c in time window t
- $N(E_{Between})_{t,c}$ measures the total number of edges between article nodes of different clusters for cluster c in time window t . It's defined as $N(E_{Between})_{t,c} = \sum_{e=(v_i,v_j)} (1_{cluster\{(v_i)=c\}} 1_{cluster\{(v_j) \neq c\}} + 1_{cluster\{(v_i) \neq c\}} 1_{cluster\{(v_j)=c\}})$



Evolution of Relative Number of Edges within Clusters



Evolution of Relative Number of Edges Between Clusters



2.3 Internal and External Edge Ratio

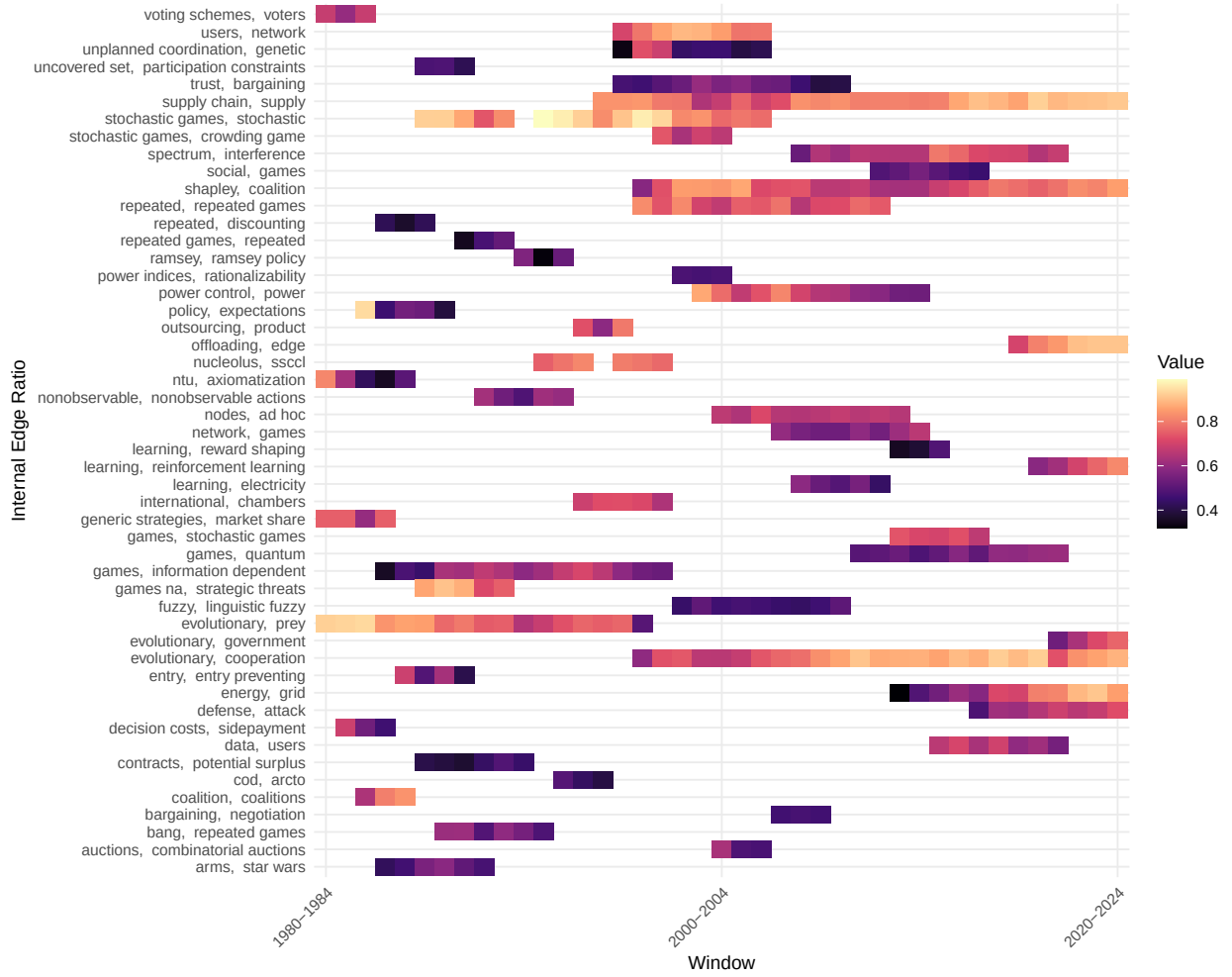
- Internal Edge Ratio:

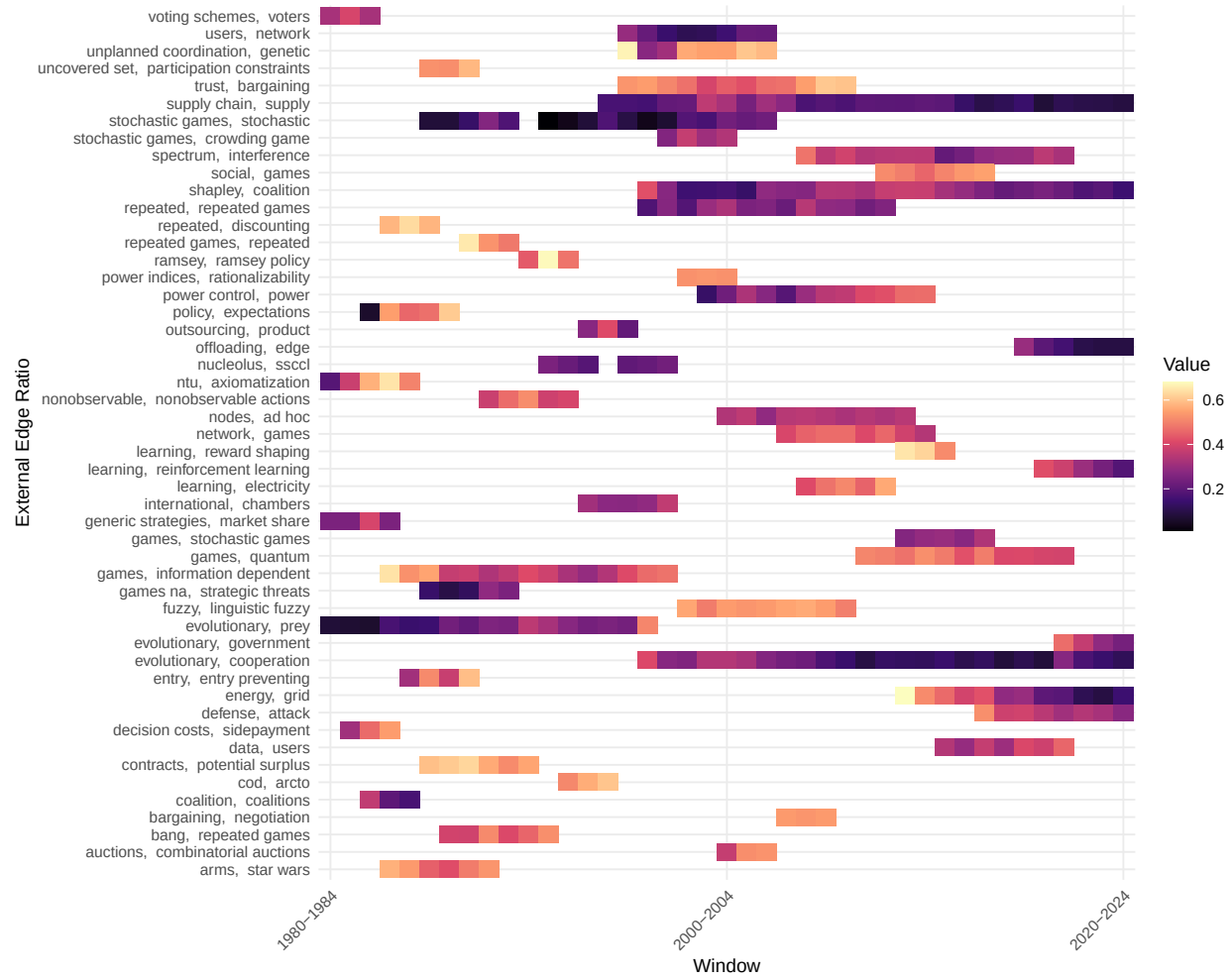
$$IR_{t,c} = \frac{N(E_{within})_{t,c}}{N(E)_{t,c}} \quad (5)$$

- External Edge Ratio:

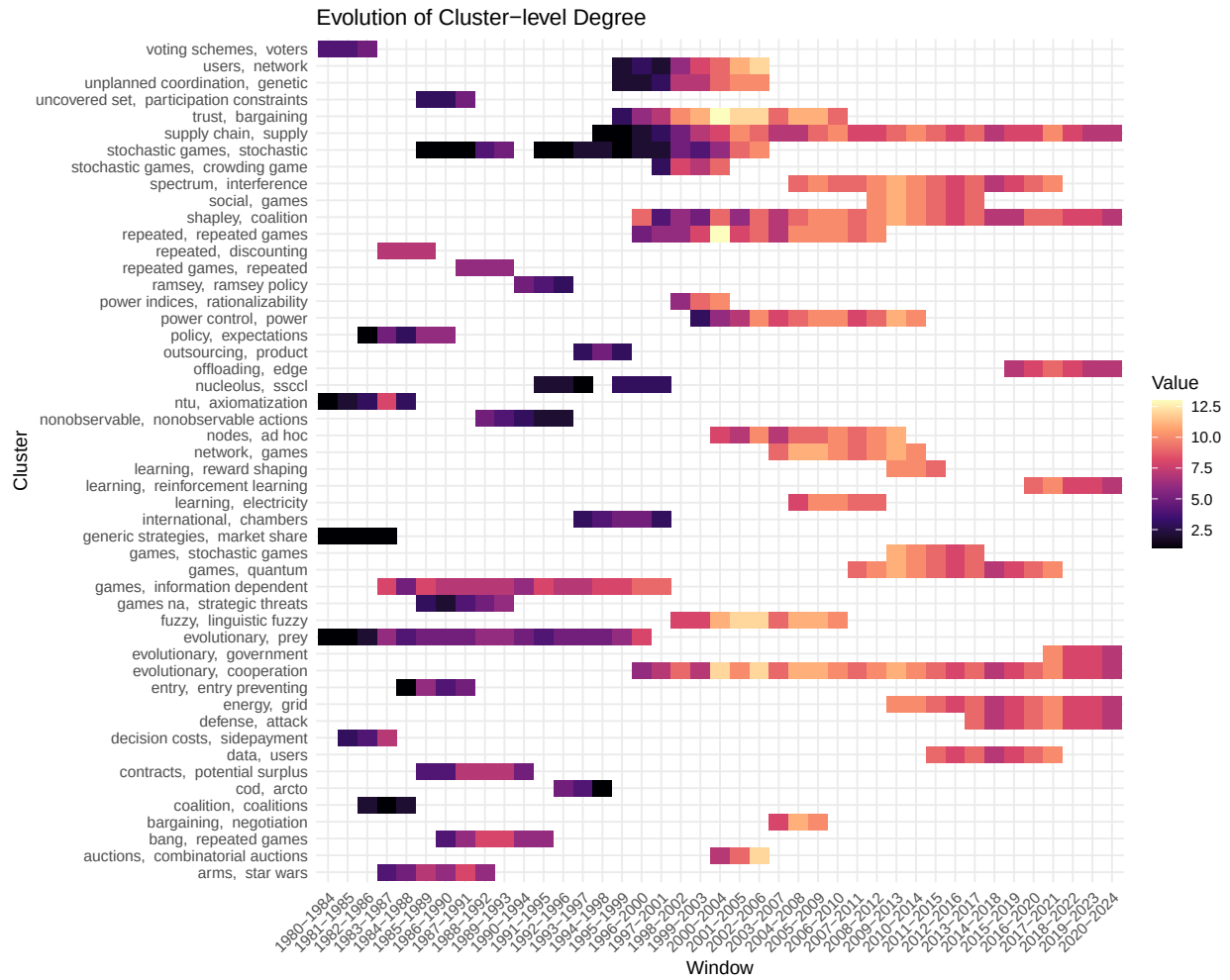
$$ER_{t,c} = \frac{N(E_{between})_{t,c}}{N(E)_{t,c}} \quad (6)$$

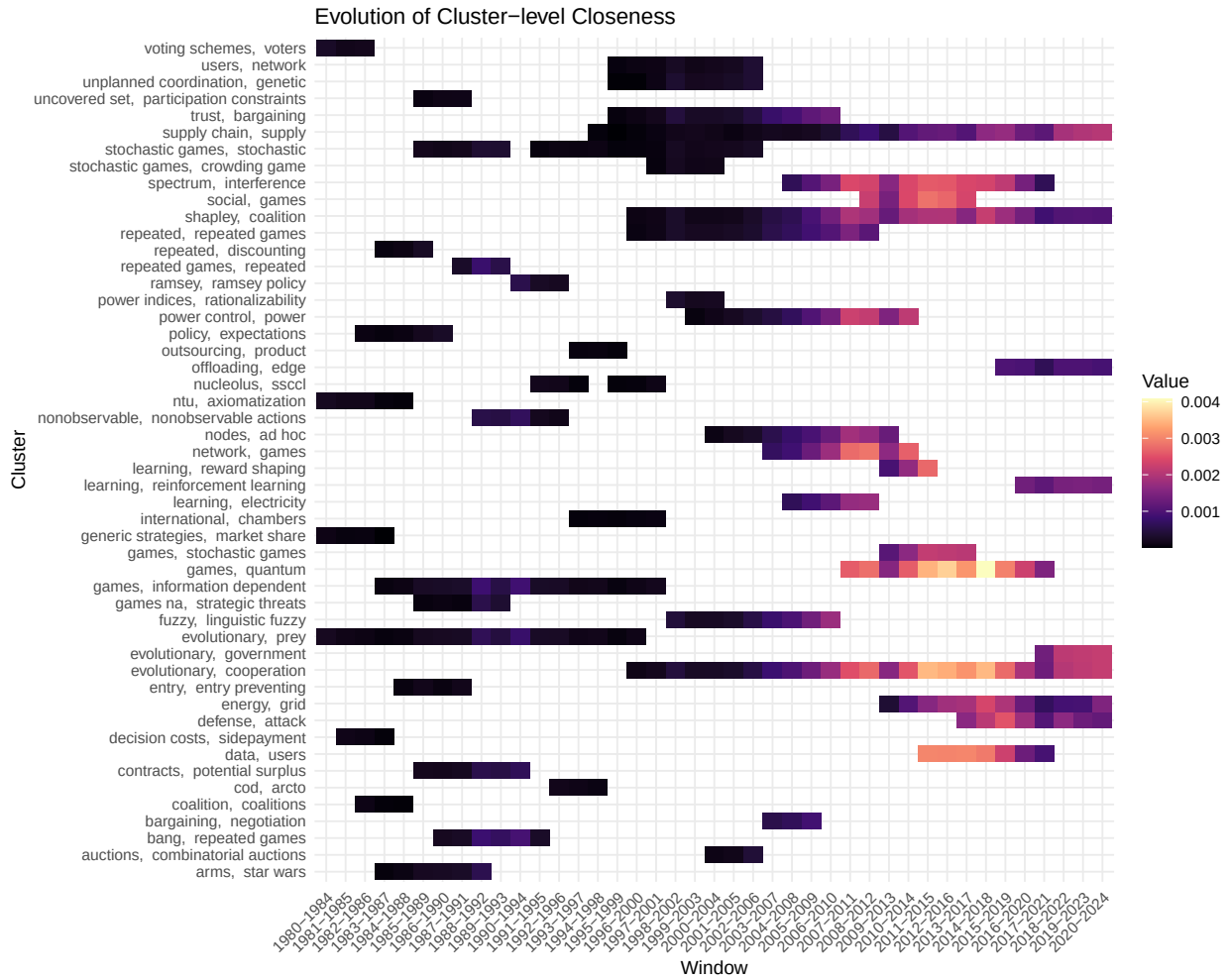
with $N(E_{within})_{t,c}$, $N(E_{between})_{t,c}$ and $N(E)_{t,c}$ defined as before. The internal and external ratios therefore measure the ratio of edges within (between) fields relative to the total number of edges of a field. Edges are derived from article-level nodes.

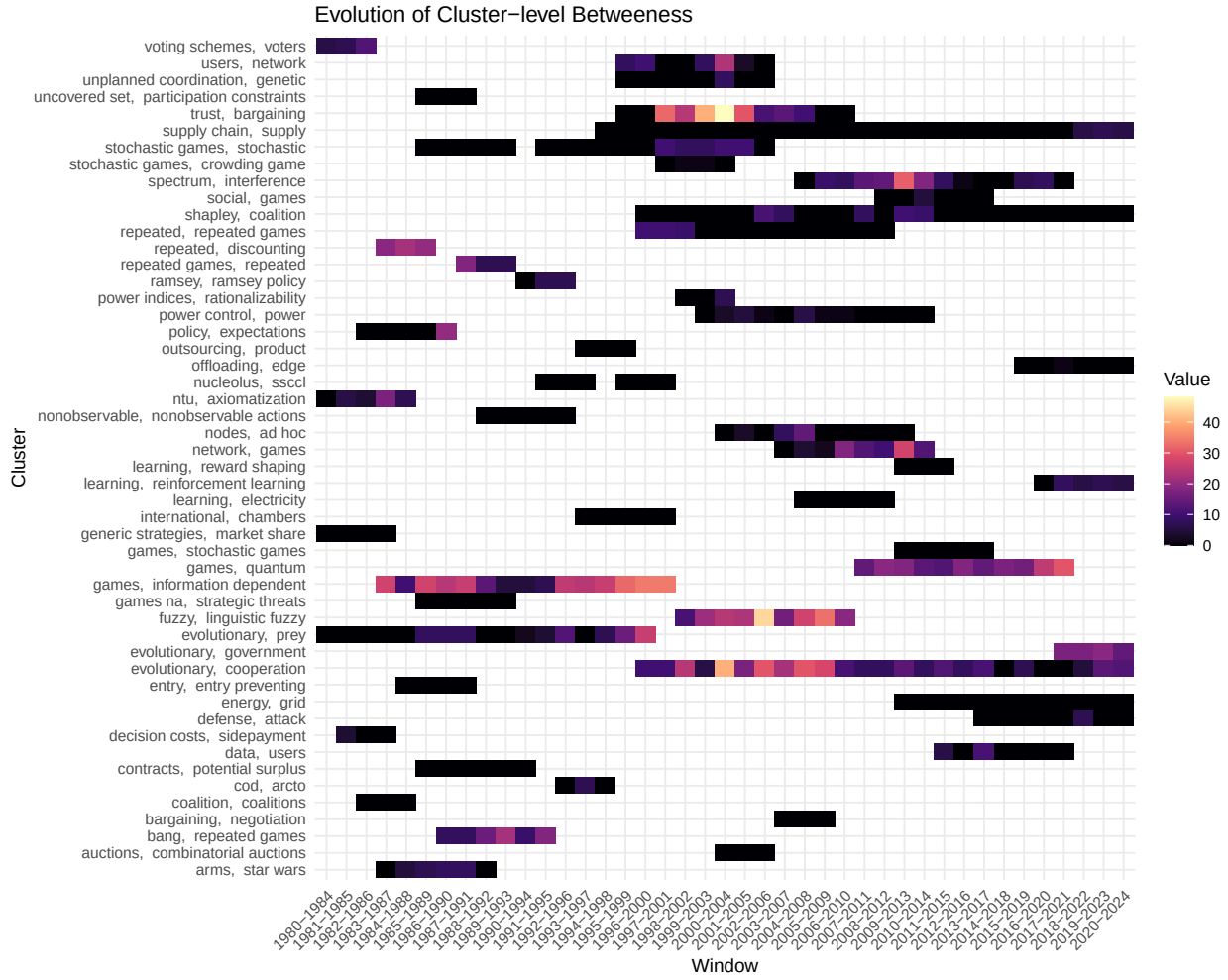




2.4 Network Centrality Metrics (Degree, Closeness and Betweenness)

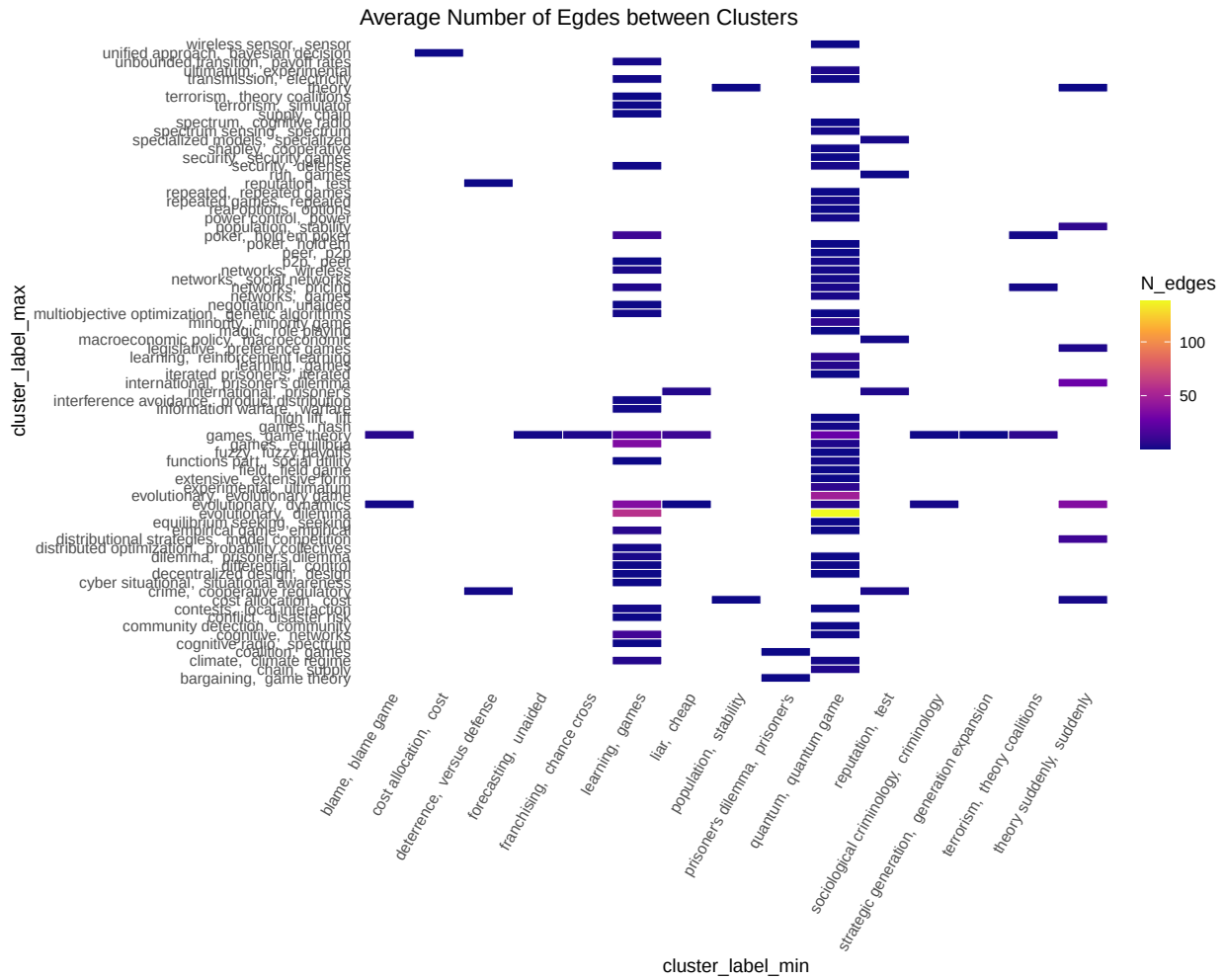


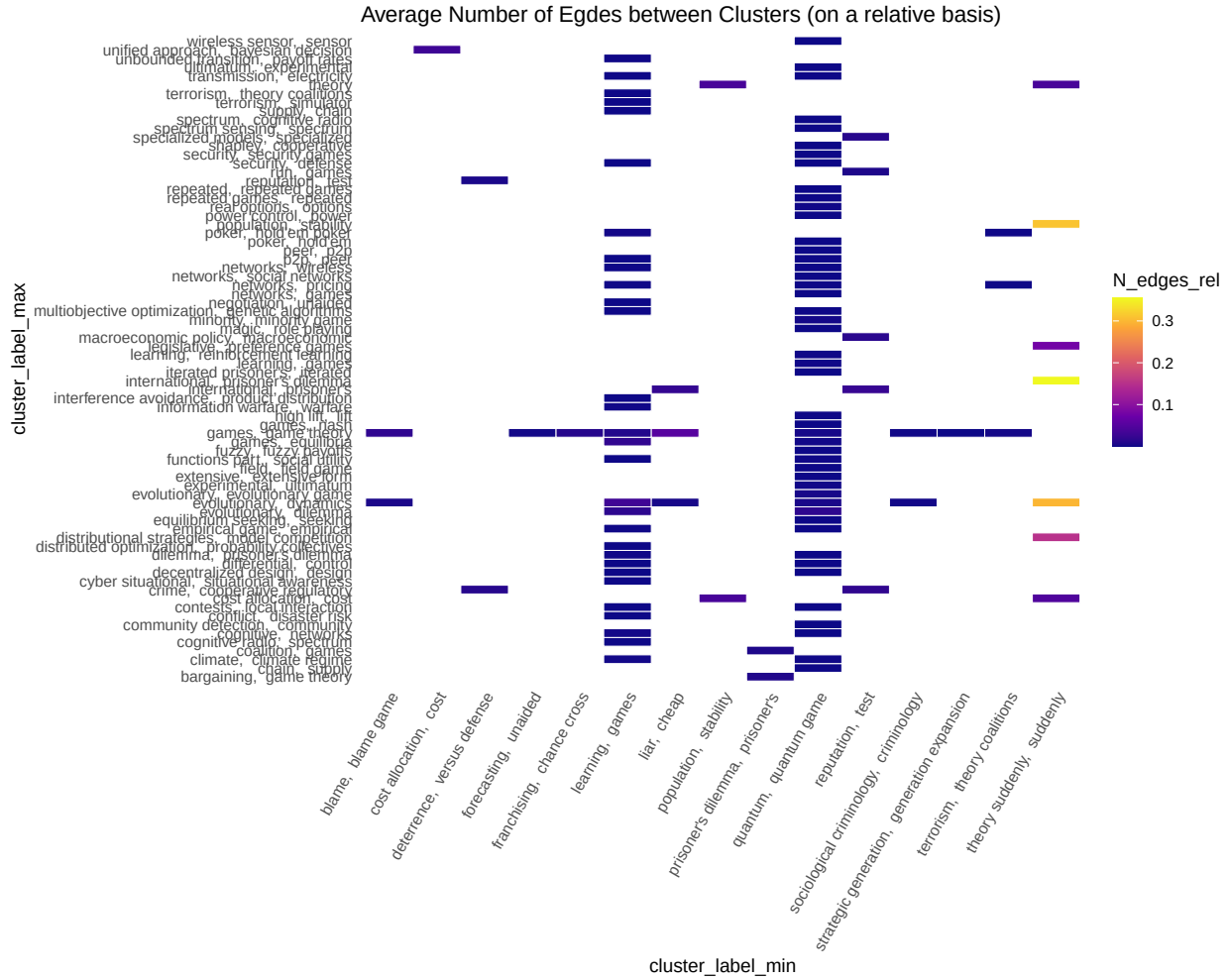




3. Field-Field Connectivity

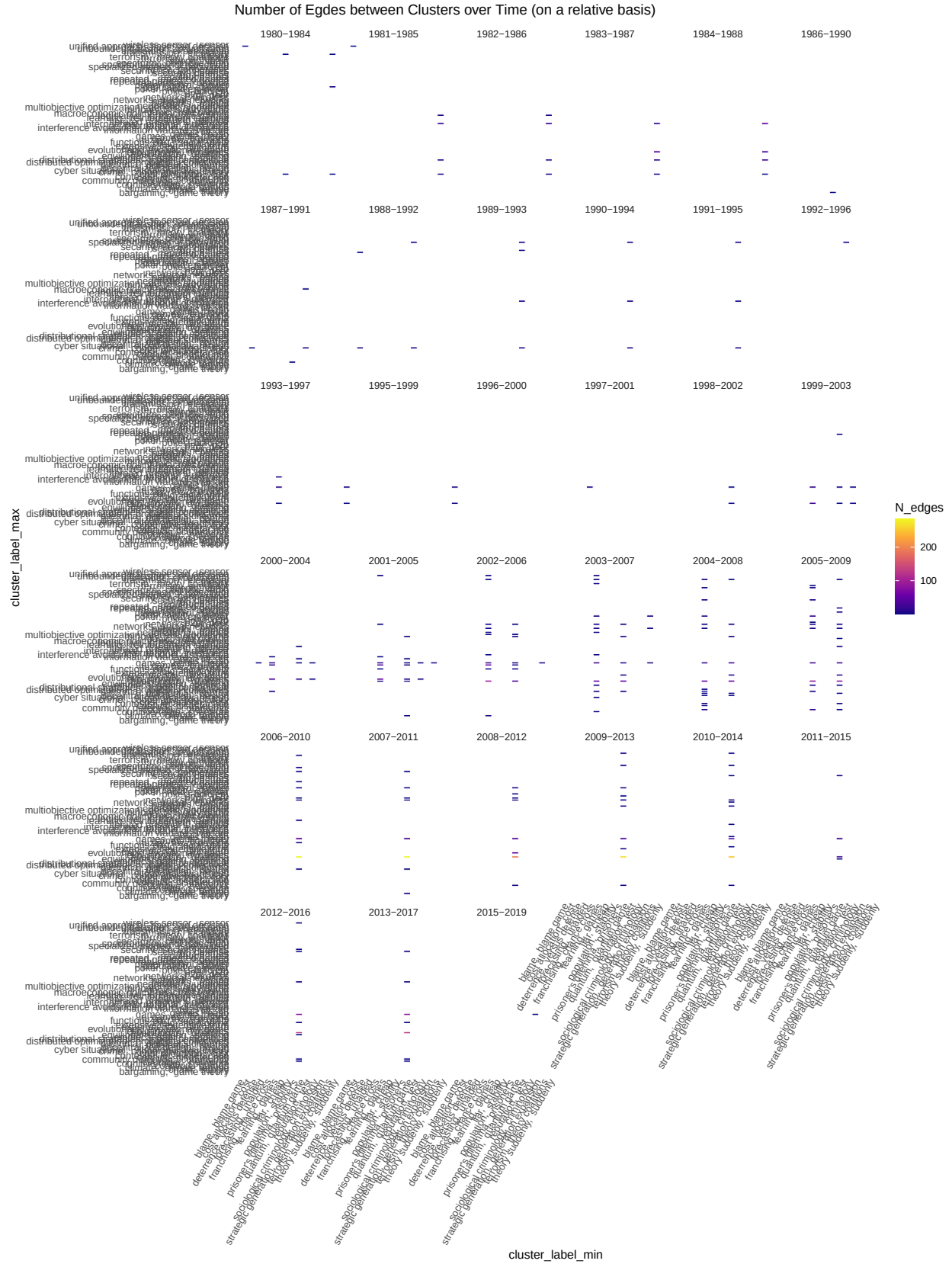
3.1 Average Connectivity





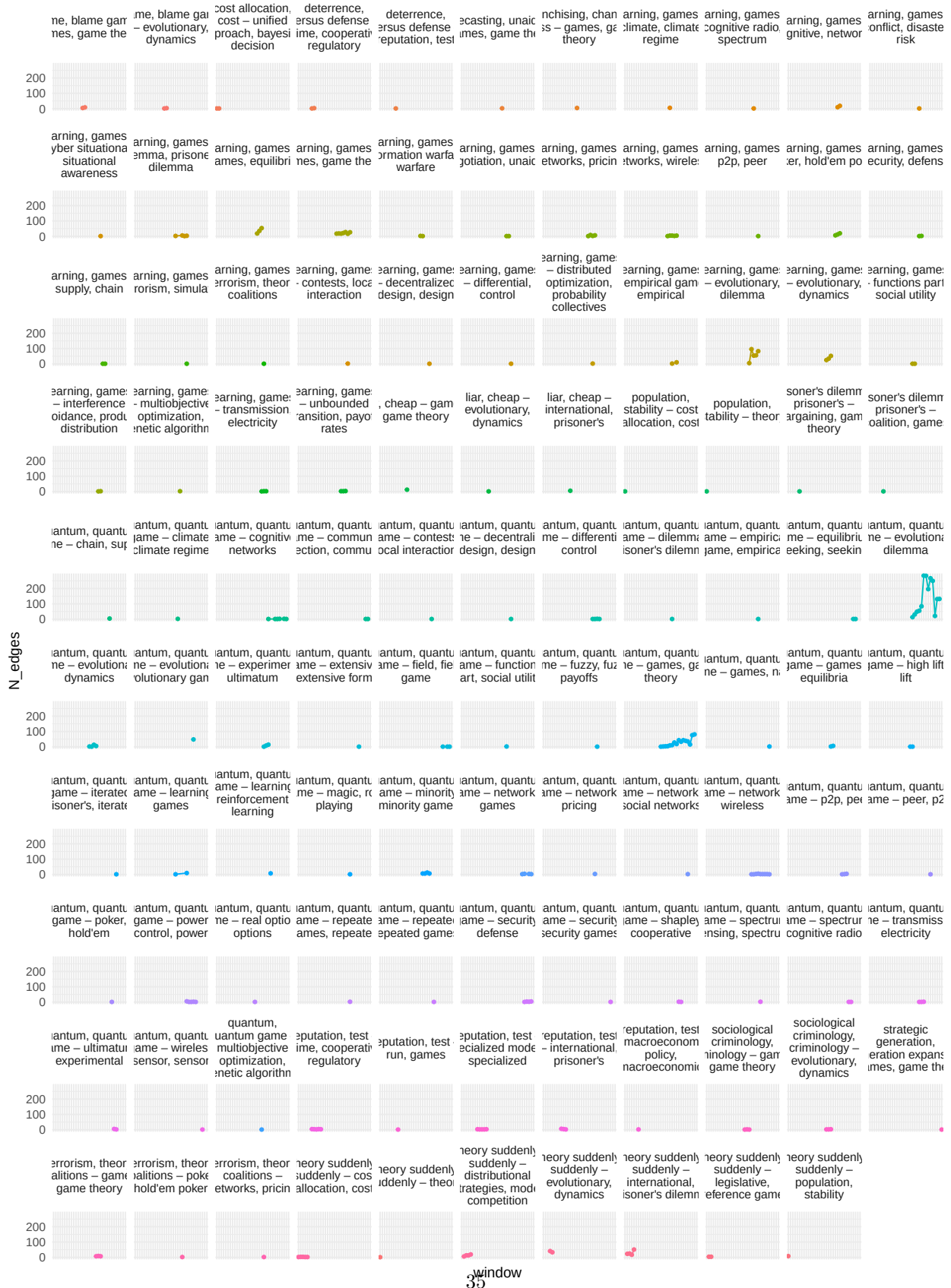
3.2.1 Heatmaps





3.2.2 Time Series

Number of Edges between Clusters over Time



Number of Edges between Clusters over Time (on a relative basis)

